

Unit III

FILES TEMPLATES AND EXCEPTION HANDLING

Course Objective – COB1: To understand the digital circuits using simplified Boolean Function

Course Outcome - CO1 : Design the digital circuits using simplified Boolean Functions.

Bloom's Taxonomy Level:

Part A

S.No.	Question	BTL*
1.	What is an Exception? Conditions responsible for creating errors during the execution of a program are known as Exceptions. These errors can interrupt the execution of the program and if the program can't handle these exceptions then OS handles them and the program is terminated abruptly.	
2.	Explain Synchronous Exceptions and Asynchronous Exceptions? Synchronous exceptions are the exceptions that occur at a particular instruction. They can only be originated from throw expressions and are caused due to errors like incorrect input or array out-of-index access in a program. Asynchronous exceptions are the exceptions that create errors that are not controllable by the program. For example-hardware malfunctions, disk failure, etc.	
3.	What is Exception Handling? Conditions responsible for creating errors during the execution of a program are known as Exceptions. Handling these exceptions by either removing these conditions or by using some other operations than normal operations are known as exception handling. Exception handling is an effective means to handle the runtime errors that disrupt the normal flow of the program.	
4.	Why do we need exception Handling? We use Exception Handling for the following reasons: 1. Separate Error code from Normal code to help us understand errors easily. 2. Functions/Methods can be handled only by the exceptions they choose. The exceptions not chosen will be handled by the caller. 3. Exceptional handling allows the grouping of error types which helps in categorizing them. 4. It makes the program's error type easy to understand.	
5.	How to implement exception handling in C++? C++ supports exception handling. It is implemented by try{ } and catch(){ } statements.	

	• The try statement allows you to define a block of code to be tested for errors while it is being executed. • The throw keyword throws an exception when a problem is detected, which lets us create a custom error. • The catch statement allows you to define a block of code to be executed if an error occurs in the try block.	
6.	What is the use of terminate() in C++? The terminate() function is used to abort the program by default rather than throwing it for a catch. The terminate() function can be used for the exceptions which can't be handled instead we just abort them. The terminate() functions calls the terminate_handler which by default calls the abort() function.	
7.	What is the use of unexpected() in C++? The unexpected() function is called when the exception thrown by a function is a type not listed in the exception specification for the function. The unexpected() will call the unexpected_handler that by default calls the terminate().	
8.	What will happen if an exception is thrown but not caught anywhere? When an exception is thrown but not caught anywhere, the program will terminate abnormally.	
9.	Explain the concept of Rethrowing exceptions? A rethrowing exception is a term used when we throw the exception again from one catch block to another.	
10.	Write a note on File. A file is a collection of related information normally representing programs; both source and object forms and data. Data may be numeric, alphabetic or alphanumeric. A file can also be defined as a sequence of bits, bytes, lines or records whose meaning is defined by the programmer. Operations such as create, open, read, write and close are performed on files	
11.	Define ifstream & fstream. ifstream: It is used for handling input files. It contains open() with default input mode and inherits get(), getline(), read(), seekg(), tellg() functions from istream. fstream: Used for handling files on which both i/o operations can be performed. It supports simultaneous i/o operations. It contains open() with default input mode and inherits all the functions from istream and ostream classes through iostream.	
12.	List the file pointer control functions. a) seekg() – moves get file pointer to a specific location	

	b) seekp() – moves put file pointer to a specific location c) tellg() – returns the current position of the get pointer d) tellp() – returns the current position of the put pointer	
13.	What are the types of file accessing? a) Sequential access This type of file is to be accessed sequentially that is to access a particular data all the preceding data items have to be read and discarded. b) Random access This type of file allows access to the specific data directly with out accessing its preceding data items	
14.	What is a stream? Stream is a series of bytes, which act either as a source from which input data can be extracted or as a destination to which the output can be sent. The source stream provides data to the program called te input stream and the destination stream that receives data from the program is called the output stream.	
15.	What are manipulators? Manipulators are special functions that are specifically designed to modify the working of a stream. They can be embedded in the I/O statements to modify the form parameters of a stream.	
16.	How many classes are used for handling files? -ifstream – for handling input files. -ofstream – for handling output files. -fstream – for handling files on which both input and output can be performed.	
17.	What are the steps involved in manipulating a file? . Name the file on the disk . Open the file to get the file pointer. . Process the file. (Read / Write) . Check for errors while processing. . Close the file after its complete usage.	
18.	What do you mean by sequential access?	

	A sequential file has to be accessed sequentially; to access the particular data in the file all the preceding data items have to be read and discarded. For example a file on a tape must be accessed sequentially.	
19.	What do you mean by random access? A random file allows access to the specific data without the need for accessing its preceding data items. However, it can be accessed sequentially. For example, a file on a hard disk or floppy disk can be accessed either sequentially or randomly.	
20.	List five common examples of exceptions. Common examples of exceptions are divide by zero, access to an array outside of its bounds, running out of memory, running out of disk space, abnormal program termination.	

Part B

S.No.	Question	BTL *
1.	What are streams? Why are they useful?(16)	
2.	Explain about formatted and unformatted IO with suitable examples.(16)	
3.	What is manipulator? Difference between manipulators and ios function.(16)	
4.	Explain the process of open, read, write and close files.(16)	
5.	Explain the role of seekg(), seekp(), tellg(), tellp() function in the process of random access in a binary file.(16)	
6.	Explain the function template.(16)	
7.	Explain the class template.(16)	
8.	What is the need for exception handling? (16)	
9.	When do we need multiple catch blocks for a single try block? Give an example.(16)	